Device Art

Course Code: PSAM 2230 | CRN 15895

Institution: The New School | Parsons School of Design | School of Art, Media, and Technology (AMT)

Term: Fall 2026 (August 26, 2026 – December 09, 2026)

Meeting Time: Wednesday Evenings, 19:00 – 21:40 (7:00pm – 9:40pm)

Location: In-Person (Parsons Making Center / Classroom TBD)

Instructor: Ariel Churi

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Office Hours: By appointment

Course Description

Device Art is a platform for students to invent new works of art that do not distinguish the object from a tool, the mechanism from the concept; through playfulness these devices are caught between definitions of art, design, and engineering. In this course devices are the artwork themselves; the mechanisms of the piece become part of the concept behind the work and the very essence of “making” and “play” is embedded in each of the pieces. Device Art can break the mold of art commercialization, propelled by the new forms of manufacturing and the ease of prototyping as production tools become more accessible. These artworks are easily replicable and can become commercialized, and even display features of playful utility for everyday life. Students will learn new methods for prototyping by combining new digital fabrication techniques as well as commonly used fabrication processes including repurposing readymades. They will explore their own artistic concepts and develop a deeper understanding of how their ideas can be reproduced. Device Art will challenge students to rethink their ideas of what art can become, and where the threshold lies between playfulness, utility, and conceptual inquiry. They will develop basic electronics and physical computing skills, such as microcontroller programming and circuitry design. They will engage in fabrication experiments to rethink and envision new devices that can facilitate invention from an artistic perspective, gaining new skills in device design, computing media, and device fabrication. They will use physical computing methods to expand on the field of open source hardware, learn interaction design methods and rapid prototyping tools for 3D printing. This course will provide platform to enhance the design process of material experimentation and propel functional and accessible models of device art.

- **Open to**: All university undergraduate degree students. Some seats have been reserved for BFA Design & Technology majors.

Learning Outcomes

By the successful completion of this course, students will be able to:

1. **Integrate skills in physical computing and fabrication** to create unique, multi-sensory interactive works.

2. **Confidently understand the fundamental physical phenomena** underlying electronics and physical computing, including voltage, current, resistance, power, and signal integrity.

3. **Prototype and document circuit designs** composed of thoughtfully selected electronic components, sensors, actuators (motors/servos), and OLED displays.

4. **Program microcontroller-driven projects** using the Raspberry Pi Pico and CircuitPython.

5. **Break down a high-level creative idea** into its constituent hardware and software subsystems.

6. **Apply foundational digital fabrication techniques**, including 3D modeling (Fusion 360) for FDM 3D printing and 2D vector preparation (Adobe Illustrator) for laser cutting.

7. **Safely work with electricity**, batteries, power supplies, a wide variety of electrical components, and digital fabrication tools.

8. **Participate in nuanced discussions and critiques** about the relationship between electronic/physical media and current social issues by drawing context from relevant history, critical theory, and contemporary practice.

9. **Build a final interactive capstone project** that integrates varying media and sensory themes in a creative practice of their own choosing.

Required Materials & Hardware Kit

Students are required to have a physical computing prototyping kit.

- **Instructor Kit Bundle ($50)**: A pre-assembled wholesale bundle available directly from the instructor (payable via Venmo, Zelle, PayPal, or cash). Kits will be distributed in **Week 2**.
- **Independent Sourcing**: A verified parts list with vendor links (Adafruit/Amazon) is provided on Canvas for students who choose to source their own parts.

Final Grade Calculation

- **1/3** Projects (Project 1, Project 2, Final Project)
- **1/3** Weekly Assignments & Technical Labs
- **1/3** Class Participation, Attendance & Critiques

Course Schedule

(Weekly topics, studio labs, and assignment briefs are managed by the instructor and published to Canvas).

Phase 1: Foundations & Electronics (Weeks 01–05)

- **Week 01 (Aug 26)**: Introductions & What is Device Art?
- Navigating the terminal
- Create files and folders
- Write code in Nano
- Run code in Python
- Print to the screen
- Variables and mathematical operations
- **Week 02 (Sep 02)**: Programming & Materials Distribution (Kits handed out in class)
- Thinking like a computer
- Conditional and loops

- **Week 03 (Sep 09)**: Electricity Fundamentals & powering you breadboard
- Create a circuit
- What is electricity?
- Volts, Amps, & Ohms
- Paper prototyping in Miro
- **Week 04 (Sep 16)**: Microcontroller Programming & Circuitry
- **Week 05 (Sep 23)**: Prototyping & Sensor Labs
- **Week 06 (Sep 30)**: Project 1 Reviews & Critiques

Phase 2: Intermediate Prototyping & Mechanisms (Weeks 07–10)

- **Week 07 (Oct 07)**: Advanced Circuitry & Responsive Behaviors
- **Week 08 (Oct 14)**: Mechanical Movement & Fabrication
- **Week 09 (Oct 21)**: Project 2 Midterm Reviews & Critiques
- **Week 10 (Oct 28)**: System Architecture & Final Project Planning
- **Week 11 (Nov 04)**: Working Prototype 1 Bench Testing

Phase 3: Final Device Fabrication & Exhibition (Weeks 12–15)

- **Week 12 (Nov 11)**: Enclosure Fabrication & Assembly
- **Week 13 (Nov 18)**: Documentation Staging & Exhibition Prep

(Nov 25: Thanksgiving Eve — No Class)

- **Week 14 (Dec 02)**: Refirement and Testing
- **Week 15 (Dec 09)**: Final Exhibition & Critique

UNIVERSITY POLICY & RESOURCES

Resources

The university provides many resources to help students achieve academic and artistic excellence. These resources include:

The University (and associated) Libraries

[The University (and associated) Libraries](https://library.newschool.edu/)

The University Learning Center

[The University Learning Center](https://www.newschool.edu/learning-center/)

University Disabilities Services

In keeping with the university's policy of providing equal access for students with disabilities, any student with a disability who needs academic accommodations must contact SDS. There are several ways for students to contact the office: via email at StudentDisability@newschool.edu, through the Starfish service catalog, or by calling the office at 212.229.5626. A self-ID form can also be completed on the SDS webpage at www.newschool.edu/student-disability-services. Once you contact the office, SDS staff will arrange an intake appointment to discuss your concerns and, if appropriate, provide you with accommodation notices to give to me. Please note that faculty will not work unilaterally with students to provide accommodations. If you inform me of a disability but do not provide any official notification, I must refer you to SDS.

Making Center

The Making Center is a constellation of shops, labs, and open workspaces that are situated across the New School to help students express their ideas in a variety of materials and methods. We have resources to help support woodworking, metalworking, ceramics and pottery work, photography and film, textiles, printmaking, 3D printing, manual and CNC machining, and more. A staff of technicians and student workers provide expertise and maintain the different shops and labs. Safety is a primary concern, so each area has policies for access, training, and etiquette with which students and faculty should be familiar. Many areas require specific orientations or training before access is granted.

Health and Wellness

[Health and Wellness](https://www.newschool.edu/student-services/health-wellness/): additional services and support available to New School students.

Learning Together/Community Agreement

At the beginning of the semester students and faculty will work together to establish a common culture/approach for learning together that could take the form of Community Agreements. As you prepare for this process, it is important for you to review the following statements:

- [**CURRICULAR JUSTICE ACKNOWLEDGMENT STATEMENT**](https://docs.google.com/document/d/14V-8ZIv2iLAY23y2WNBeHAHvZhaUU00bsG9txSAhl4Y/edit?usp=sharing)
- [**STATEMENT ON DIVERSITY, EQUITY & INCLUSION**](https://docs.google.com/document/d/1kuMJ6carMfWwjPEiSnm7FUKNn2U0hOwG3T6r0ZNMpac/edit?usp=sharing)
- [**PRONOUNS STATEMENT**](https://docs.google.com/document/d/1G2MvyMR9jLRjMuM_FDHVZp6LIpWn_ycGzny9UZyRYoc/edit?usp=sharing)

Grading Standards

Undergraduate

A student's final grades and GPA are calculated using a 4.0 scale.

A [4.0] Work of exceptional quality, which often goes beyond the stated goals of the course (95-100%)

A- [3.7] Work of very high quality (90% - <95%)

B+ [3.3] Work of high quality that indicates higher than average abilities (87% - <90%)

B [3.0] Very good work that satisfies the goals of the course (83% - <87%)

B- [2.7] Good work (80% - <83%)

C+ [2.3] Above-average work (77% - <80%)

C [2.0] Average work that indicates an understanding of the course material; passable (73% - <77%); Satisfactory completion of a course is considered to be a grade of C or higher.

C- [1.7] Passing work but below good academic standing (70% - <73%)

D [1.0] Below-average work that indicates a student does not fully understand the assignments (60% - <70%); Probation level though passing for credit

F [0.0] Failure, no credit (0% - <60%)

GM Grade missing for an individual

Grade of W

The grade of W may be issued by the Office of the Registrar to a student who officially withdraws from a course within the applicable deadline. There is no academic penalty, but the grade will appear on the student transcript.

Grades of Incomplete

The grade of I, or temporary incomplete, may be granted to a student under unusual and extenuating circumstances, such as when the student's academic life is interrupted by a medical or personal emergency. This mark is not given automatically but only upon the student's request and at the discretion of the instructor. A Request for Incomplete form must be completed and signed by the student, instructor and program director. The time allowed for completion of the work and removal of the "I" mark will be set by the instructor with the following limitations:

Undergraduate students: Work must be completed no later than the seventh week of the following fall semester for spring or summer term incompletes and no later than the seventh week of the following spring semester for fall term incompletes. Grades of "I" not revised in the prescribed time will be recorded as a final grade of "F" by the Registrar's Office.

College, School, Program and Class Policies

A comprehensive overview of policy may be found under [Policies: A to Z](https://www.newschool.edu/student-conduct/policies/). Students are also encouraged to consult the [Academic Catalog for Parsons](https://www.newschool.edu/parsons/academic-catalog/).

Canvas

Use of Canvas may be an important resource for this class. Students should check it for announcements before coming to class each week.

Electronic Devices

The use of electronic devices (phones, tablets, laptops, cameras, etc.) is permitted when the device is being used in relation to the course's work. All other uses are prohibited in the classroom and devices should be turned off before class starts.

Responsibility

Students are responsible for all assignments, even if they are absent. Late assignments, failure to complete the assignments for class discussion and/or critique, and lack of preparedness for in-class discussions, presentations and/or critiques will jeopardize your successful completion of this course.

Active Participation and Attendance

Class participation is an essential part of class and includes: keeping up with reading, assignments, projects, contributing meaningfully to class discussions, active participation in group work, and attending synchronous sessions regularly and on time.

Parsons' attendance guidelines were developed to encourage students' success in all aspects of their academic programs. Full participation is essential to the successful completion of coursework and enhances the quality of the educational experience for all, particularly in courses where group work is integral; thus, Parsons promotes high levels of attendance. Students are expected to attend classes regularly and promptly and in compliance with the standards stated in this course syllabus.

While attendance is just one aspect of active participation, absence from a significant portion of class time may prevent the successful attainment of course objectives. A significant portion of class time is generally defined as the equivalent of three weeks, or 20%, of class time. Lateness or early departure from class may be recorded as one full absence. Students may be asked to withdraw from a course if habitual absenteeism or tardiness has a negative impact on the class environment.

Using AI and Generative Tools

Please note that the use of Artificial Intelligence and generative text and image tools are addressed in [the New School statement on Student Academic Integrity](https://

www.newschool.edu/student-conduct/academic-integrity/). As a general rule please consider the following with regards to AI:

- Do not submit any final work - writing or images - generated by AI tools without written permission from your faculty.
- AI tools deliver results that have been synthesized and averaged from many non citable sources. Misrepresentations are easy to miss.
- The AI tech space is rapidly changing and will continue to be contested. Be discerning in how you use these tools.
- Remember why you are at Parsons - to invest in yourself as a creative. Explore these tools with curiosity and criticality, rather than dependency.

Faculty Specific policy about the use of AI and generative tools:

Students may use AI as a learning assistant to clarify code or brainstorm, but **no final work (text, code, or images) generated by AI may be submitted** as your own without written permission. All code submitted must be written and understood by you. You will be expected to explain your code and problem-solving process during critiques and reviews.

Academic Honesty and Integrity

Compromising your academic integrity may lead to serious consequences, including (but not limited to) one or more of the following: failure of the assignment, failure of the course, academic warning, disciplinary probation, suspension from the university, or dismissal from the university.

Students are responsible for understanding the University's policy on academic honesty and integrity and must make use of proper citations of sources for writing papers, creating, presenting, and performing their work, taking examinations, and doing research. It is the responsibility of students to learn the procedures specific to their discipline for correctly and appropriately differentiating their own work from that of others. The full text of the policy, including adjudication procedures, is found on the university website under [Policies: A to Z](https://www.newschool.edu/student-conduct/policies/). Resources regarding what plagiarism is and how to avoid it can be found on the [Learning Center's website](https://www.newschool.edu/learning-center/).

The New School views "academic honesty and integrity" as the duty of every member of an academic community to claim authorship for his or her own work and only for that work, and to recognize the contributions of others accurately and completely. This obligation is fundamental to the integrity of intellectual debate, and creative and academic pursuits. Academic honesty and integrity includes accurate use of quotations, as well as appropriate and explicit citation of sources in instances of paraphrasing and describing ideas, or reporting on research findings or any aspect of the work of others (including that of faculty members and other students). Academic dishonesty results from infractions of this "accurate use". The standards of academic honesty and integrity, and citation of sources, apply to all forms of academic work, including submissions of drafts of final papers or projects. All members of the University community are expected to conduct themselves in accord with the standards of academic honesty and integrity. Please see the complete policy in the Parsons Catalog.

Intellectual Property Rights

The New School (the "university") seeks to encourage creativity and invention among its faculty members and students. In doing so, the University affirms its traditional commitment to the personal ownership by its faculty members and students of Intellectual Property Rights in works they create. The complete policy governing Intellectual Property Rights may be seen on the university website, on the [Provost's page](https://www.newschool.edu/provost/policies/).

Student Course Ratings (Course Evaluations)

During the last two weeks of the semester, students are asked to provide feedback for each of their courses through an online survey. They cannot view grades until providing feedback or officially declining to do so. Course evaluations are a vital space where students can speak about the learning experience. It is an important process which provides valuable data about the successful delivery and support of a course or topic to both the faculty and administrators. Instructors rely on course rating surveys for feedback on the course and teaching methods, so they can understand what aspects of the class are most successful in teaching students, and what aspects might be improved or changed in future. Without this information, it can be difficult for an instructor to reflect upon and improve teaching methods and course design. In addition, program/department chairs and other administrators review course surveys. Instructions are available online [here](https://www.newschool.edu/registrar/course-evaluations/).